Hypermobile Ehlers-Danlos Syndrome (hEDS)

A Patient Guide for Optometrists & Ophthalmologists

WHAT IS hEDS? hEDS is a heritable disorder of connective tissue, the structural 'glue' of the body, causing joint instability, The eye is largely collagen: the cornea (~70% type I collagen), sclera (~90% type I collagen), lens zonules, and vitreous are all affected by faulty connective tissue in hEDS. Ocular manifestations are common and frequently overlooked. Severity varies widely but often includes dry eye, convergence insufficiency, myopia, and increased surgical risk.

~1 in 500 people affected	Avg. 10+ years to diagnosis	3:1 to 4:1 diagnosed are female	No cure: management-focused

HOW hEDS AFFECTS THE VISUAL SYSTEM:

Patient has checked applicable symptoms

Refractive & Structural

- ☐ Myopia (nearsightedness), often progressive
- ☐ Steeper corneal curvature / higher corneal power
- ☐ Thin or variable corneal thickness (pachymetry)
- ☐ Astigmatism
- ☐ Minor lens opacities (early onset possible)
- ☐ Vitreous abnormalities / floaters
- ☐ Axial elongation (longer eye)

Ocular Surface & Adnexa

- ☐ Dry eye / xerophthalmia (very common)
- ☐ Reduced tear film break-up time
- ☐ Eyelid laxity / floppy eyelid features
- ☐ Periorbital dark circles (visible vessels through fragile skin)
- ☐ Conjunctivochalasis (excess conjunctival tissue)
- ☐ Photosensitivity / light sensitivity
- ☐ Chemical / fragrance sensitivity affecting eyes

Motility, Pressure & Retinal

- ☐ Convergence insufficiency
- ☐ Eye strain, diplopia at near tasks
- ☐ Accommodative dysfunction
- ☐ Potentially altered IOP (thin corneas affect reading)
- ☐ Increased retinal detachment risk with high myopia
- ☐ Blue sclera (typically mild in hEDS vs. other subtypes)
- ☐ Symptoms worsen with fatigue & autonomic flares (POTS)

DO

- Perform corneal pachymetry at baseline: thin corneas affect IOP accuracy and surgical risk stratification
- Obtain corneal topography: screen for subclinical ectasia and keratoconus, which occur at higher rates in hEDS
- Assess convergence and accommodative function: convergence insufficiency is common and treatable
- Conduct dilated fundus exam with attention to peripheral retina, especially in high myopia
- Treat dry eye proactively: preservative-free artificial tears, punctal plugs, and omega-3 as clinically appropriate
- Ask about POTS and autonomic symptoms: orthostatic changes can transiently affect vision and exacerbate dry eye

DON'T

- Recommend LASIK or similar flap-based corneal refractive surgery without thorough connective tissue risk assessment: the FDA contraindicates LASIK in EDS due to risk of ectasia, poor wound healing, and globe fragility
- Rely solely on standard applanation tonometry without accounting for corneal thickness: thin or atypical corneas give falsely low IOP readings
- Dismiss eye strain, visual fatigue, or diplopia as functional or psychosomatic without convergence and accommodative testing
- Assume contact lens intolerance is behavioral: dry eye and corneal sensitivity are physiological in hEDS
- Neglect to counsel against habitual eye rubbing: corneal fragility increases ectasia risk.

CONSIDER / REFER

- Vision therapy or orthoptic exercises for symptomatic convergence insufficiency: effective in majority of cases
- Scleral lenses for dry eye and/or irregular corneal surface management: superior comfort with fragile ocular tissues
- Corneal cross-linking (CXL) evaluation if progressive ectasia or keratoconus is confirmed
- Retinal specialist referral for high myopia (>-6.00 D), peripheral degeneration, or symptomatic floaters
- Phakic IOL or PRK as alternatives if refractive correction is being considered and standard corneal surgery is contraindicated: discuss with patient and corneal specialist
- Allergy/immunology co-management if MCAS is suspected as a contributor to ocular surface disease (episodic tearing, itching, chemosis)
- Coordinate with cardiology if patient has known POTS: visual symptoms may be autonomic in origin.
- Baseline dilated fundus documentation for longitudinal monitoring in younger myopic patients

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LASIK Contraindication in hEDS: What Providers Need to Know

The FDA contraindicates LASIK in Ehlers-Danlos Syndrome. Because hEDS involves structurally abnormal collagen, the cornea may be mechanically compromised even when topography appears borderline-normal. Risks include:

- **Post-LASIK ectasia:** progressive corneal steepening and thinning following surgery, leading to irregular astigmatism and vision loss
- **Poor wound healing and refractive unpredictability:** fragile tissue does not respond predictably to ablation or healing and altered biomechanics mean outcomes are less reliable
- **Globe fragility:** rare but serious risk, particularly with flap-based approaches

If refractive surgery is desired: PRK (surface ablation) may be considered in carefully selected patients with normal corneal pachymetry (>500 µm central), no topographic irregularity, and white sclera. SMILE or phakic IOL implantation may be appropriate alternatives. Always obtain informed consent specific to connective tissue risk. Refer to a corneal specialist experienced with connective tissue disorders.

Dry Eye in hEDS: Physiological, Not Behavioral

Dry eye is the most consistently reported ocular symptom in hEDS, affecting nearly all patients in clinical studies. It is not explained by screen time or environment alone. Contributing mechanisms include:

- **Connective tissue laxity in eyelids and conjunctiva:** reduces tear distribution and increases evaporative loss
- **Conjunctivochalasis:** excess conjunctival tissue disrupts the tear meniscus and creates ocular surface irritation
- **MCAS co-activation:** mast cell mediators cause episodic eye irritation, itching, tearing, and chemosis that may not respond to standard dry eye treatment
- **Autonomic dysfunction (POTS):** abnormal autonomic regulation affects lacrimal gland function and tear production variability

Preservative-free lubricants are strongly preferred. Preserved drops can worsen already-sensitive ocular surfaces. Punctal plugs, cyclosporine drops, and scleral lenses are appropriate escalation steps.

Convergence Insufficiency: Common, Underdiagnosed, Treatable

Convergence insufficiency (CI) is among the most frequently identified ocular motility findings in hEDS. It presents as eye strain, headache, blurred or double vision at near tasks, and reading fatigue, symptoms often attributed to anxiety or fatigue in this population. Prevalence of ocular motility disorders in hEDS cohorts has been reported at over 70%.

- Perform near point of convergence (NPC) testing and positive fusional vergence (PFV) assessment at each visit
- Office-based vision therapy achieves symptom resolution in approximately 75% of patients with symptomatic CI
- Base-in prism reading glasses are appropriate when vergence therapy is insufficient or not accessible
- Screen for underlying accommodative insufficiency as a co-contributor

Clinical note: CI symptoms in hEDS may worsen during autonomic flares, menstrual cycles, or periods of high fatigue. Symptom fluctuation does not mean the diagnosis is functional. Persistent visuomotor complaints including convergence insufficiency, smooth pursuit deficits, and saccadic dysfunction have in some cases been the finding that led to an hEDS diagnosis, making this a potentially diagnostically significant presentation. Frank strabismus can occur in hEDS and may warrant surgical referral, but non-surgical management should be exhausted first: strabismus surgery carries the highest ophthalmic surgical complication rate in EDS at 59%.

MCAS and the Ocular Surface: What Eye Care Providers Need to Know

Mast Cell Activation Syndrome (MCAS) co-occurs in a significant subset of hEDS patients and directly compounds ocular surface instability.

Ocular Symptoms of MCAS

- Episodic tearing without obvious trigger
- Itching, burning, chemosis
- Worsening dry eye after food, scent, or temperature
- Periorbital swelling
- Highly variable day-to-day ocular surface comfort

Triggers Relevant to Eye Care

- Preserved eye drops and topical medications
- Contrast dyes: pre-medicate if anaphylaxis history
- Fragrances in clinic (lotions, cleaning products)
- Bright lights or rapid temperature change
- NSAIDs in topical form

Clinical Note

When dry eye is highly variable, episodic, or resistant to standard treatment, consider MCAS as a co-driver. H1+H2 antihistamines may reduce ocular surface instability independently of standard dry eye therapy. Refer to Allergy/Immunology familiar with MCAS if suspected.

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Surgical Risk in hEDS: What Every Surgical Subspecialty Needs to Know

In a retrospective cohort of 467 EDS patients who underwent ophthalmic surgery, 24% required surgery and nearly 50% of those experienced at least one complication. Complication rates by procedure type included: strabismus surgery (59%), cataract/lens (33%), retinal (33%), glaucoma laser (25%), oculoplastics (23%), and refractive (23%). The majority were not yet diagnosed with EDS at the time of surgery.

- **Suture management:** tissue fragility requires gentler handling; sutures may not hold as expected
- **Wound healing:** healing is slower and less predictable; patients may need extended post-op observation
- **Anesthesia:** inform anesthesia team of connective tissue diagnosis; positioning and airway management may require modification
- **Corticosteroid drops post-op:** monitor IOP carefully; steroid response may be exaggerated or irregular
- **Informed consent:** document hEDS-specific surgical risks explicitly

COMMON MISDIAGNOSES IN hEDS PATIENTS PRESENTING TO EYE CARE

Often Diagnosed As	Consider instead/Also	Key Differentiator
Functional vision complaints / anxiety	Convergence insufficiency, accommodative dysfunction	Near point of convergence testing; symptom pattern linked to near tasks and fatigue
Simple dry eye (MGD, environmental)	hEDS-related dry eye with MCAS co-activation	Episodic pattern; triggers (food, scent, temp); poor response to standard treatment
Contact lens intolerance (behavioral)	Corneal fragility, dry eye, preserved drop sensitivity	Preserved-drop sensitivity; scleral lenses often well-tolerated where standard lenses fail
Idiopathic progressive myopia	hEDS-associated axial elongation and connective tissue laxity	Connective tissue history; steeper corneas; vitreous changes; onset in adolescence
Functional diplopia / psychosomatic	Convergence insufficiency, POTS-related visual symptoms	Positional component; orthostatic assessment; CI confirmed on vergence testing
Glaucoma suspect (low IOP reading)	Falsely low IOP due to thin corneas; true IOP may be higher	Central corneal thickness (CCT): thin corneas underestimate IOP on standard applanation tonometry; non-contact or corrected tonometry methods give more accurate readings

MY CURRENT MEDICATIONS & SUPPLEMENTS	WHAT HELPS:
	WHAT MAKES IT WORSE:

WHAT I NEED FROM TODAY'S APPOINTMENT

My primary concern today:

Questions I have:

Medication changes:

Referrals needed:

Other:

Sources: Malfait et al. 2017 (AJMG) | Gharbiya et al. 2012 (Am J Ophthalmol) | Perez-Roustit et al. 2019 (J Fr Ophtalmol) | Kim et al. 2025 (Eye) | Asanad et al. 2022 (Front Med) | Moshirfar et al. 2021 (J Clin Med) | eyewiki.org/Ehlers-Danlos_Syndrome

This document was created to provide a focused visual reference for providers less familiar with hEDS and its ocular manifestations. It is evidence-based and designed to support efficient, informed care.

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CURRENT SYMPTOM SEVERITY: Use the Mankoski Pain Scale (pg. 4) as a general severity guide, not only for pain.
Dry eye severity and frequency:
Eye strain / convergence symptoms:
Light sensitivity:
Visual disturbance with posture change:
Additional symptoms:

MANKOSKI PAIN SCALE Use this scale when rating your symptom severity

#	What the pain is like	Typical treatment	In my own words
0	No pain.	*No medication needed.*	*"I feel completely normal."*
1	Very minor annoyance — occasional minor twinges.	*No medication needed.*	*"Hardly notice it."*
2	Minor annoyance — occasional strong twinges.	*No medication needed.*	*"Annoying but manageable."*
3	Annoying enough to be distracting.	*Mild OTC painkillers may help.*	*"Hard to ignore; affects my focus."*
4	Can be ignored if very focused, but still distracting.	*Mild OTC painkillers relieve pain for 3-4 hours.*	*"Getting in the way of tasks."*
5	Can't be ignored for more than 30 minutes.	*Mild OTC painkillers reduce pain for 3-4 hours.*	*"Stops me mid-task."*
6	Can't be ignored. Can still go to work and participate in social activities.	*Stronger prescription pain relief needed; works 3-4 hours.*	*"Present all the time. I push through."*
7	Difficult to concentrate; interferes with sleep. Can still function with effort.	*Stronger painkillers only partially effective.*	*"Hard to function. Sleep is disrupted."*
8	Physical activity severely limited. Can read/converse with effort. Nausea possible.	*Strongest painkillers minimally effective.*	*"Mostly bed-bound. May feel nauseated."*
9	Unable to speak. Crying out or moaning uncontrollably. Near delirium.	*Strongest painkillers only partially effective.*	*"Cannot communicate. Losing control."*
10	Unconscious. Pain causes passing out.	*Strongest painkillers only partially effective.*	*"Passed out or on the verge of it."*

Mankoski Pain Scale developed by Andrea Mankoski, 1995. Adapted for patient communication. Not a clinical diagnostic tool.

IMPORTANT NOTE FOR HEDS PATIENTS & PROVIDERS:

People with hEDS often have an altered pain baseline due to central sensitization:
a process in which the nervous system becomes increasingly
sensitized to pain signals over time.

A '3' for this patient may be what others feel as a '6.'
Please do not compare severity numbers to those of patients without chronic illness.

This scale helps us communicate:
it is not a measure of tolerance, willpower, or how 'bad' things really are.

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